MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER MAKEUP EXAMINATION			
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**B.Tech**	**Course Type**	**PE**
Course Name	**Elements of Earthquake Engineering**	**Course Code**	**CIE 4078**
Semester	**VII**	**Max Marks**	**50**
Date	**28/12/2024**	**Duration**	**180 minutes**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**
- **Usage of IS 1893:2016**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A.	Explain box action concept.	**2**	**CO1**	1,2,3,12, PSO2	2
1B.	Determine the natural frequency of a SDOF system as shown in **Fig 1**.	**3**	**CO2**	1,2,3,12, PSO2	3
1C.	Derive the steady state response of an underdamped SDOF system subjected to force $F_0 \sin\bar{\omega}t$.	**5**	**CO3**	1,2,3,12, PSO2	2
2A.	The static deflection of an electric generator of mass 700 kg supported by a system of four parallel springs is found to be 70 mm. If the electric generator has a rotating unbalance of 0.7 kg-m, determine the amplitude of vibration at 2500 rpm.	**5**	**CO3**	1,2,3,12, PSO2	3
2B.	A frame shown in **Fig.2** supports a rotating machine, which exerts a horizontal force at the girder level is F(t) = 400 N sin 20 Hz t. Assuming 10% of critical damping, determine steady-state amplitude of vibration and the maximum dynamic bending stress in the column Take E = 200 kN/mm^2, I = $4.5x10^6$ mm^4 and Z = 2000 mm^3 for each leg.	**5**	**CO3**	1,2,3,12, PSO2	3
3A.	A frame shown in **Fig. 3**, supports a motor of mass 150 kg. The mass of floor on which the motor is mounted is 4500kg. The motor has an unbalanced force of 10 kg-m and operates at 1000 rpm. Determine the maximum amplitude of displacement and maximum bending moments in each column. Assume 10% damping and take EI=3500 $kN\text{-}m^2$.	**5**	**CO3**	1,2,3,12, PSO2	3

3B.	An electric motor weighing 2500 N and operating at 500 rpm is mounted on 4 parallel springs of stiffness 5000 N/m each. Determine the maximum permissible unbalance in order to limit the steady state amplitude of 5mm and assume damping as 15% of critical damping.	**3**	**CO3**	1,2,3,12, PSO2	3
3C.	Explain dynamic load factor.	**2**	**CO3**	1,2,3,12, PSO2	2
4A.	Determine base shear and its distribution for all the floors using equivalent static load method for the **Fig.4**. The structure is a residential apartment with less than 200 occupants, built by ordinary RC frames on soft strength soil in Delhi. Wall panels are not present in the building. The beam dimension is 230×450mm, columns are 230×600mm and slab is 200 mm thick. Live load on slab is 3.0 kN/m^2.	**8**	**CO4**	1,2,3,12, PSO2	3
4B.	Draw strorey shear and base shear diagram for the 4A. question.	**2**	**CO4**	1,2,3,12, PSO2	2
5A.	Derive the response of an undamped SDOF system subjected to force $F_0 \sin\bar{\omega}t$.	**4**	**CO3**	1,2,3,12, PSO2	2
5B.	Explain the importance of longitudinal bars lapping location in columns with neat sketch.	**4**	**CO5**	1,2,3,12, PSO2	2
5C.	Explain beam jacketing.	**2**	**CO5**	1,2,3,12, PSO2	2

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	**2**
2	**3**
3	**29**
4	**10**
5	**6**

BL	Total Marks
2	**21**
3	**29**
4	
5	

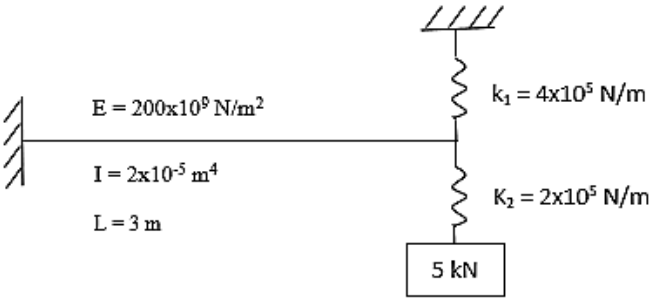


Fig. 1

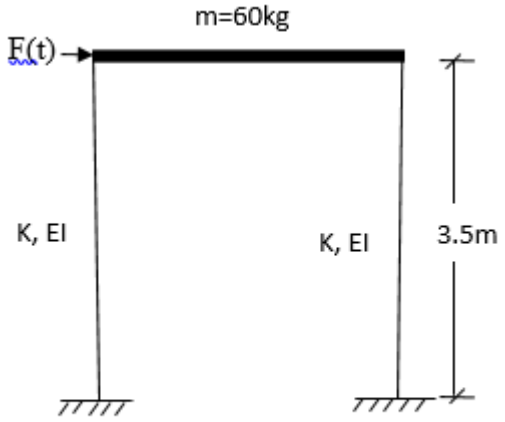


Fig. 2

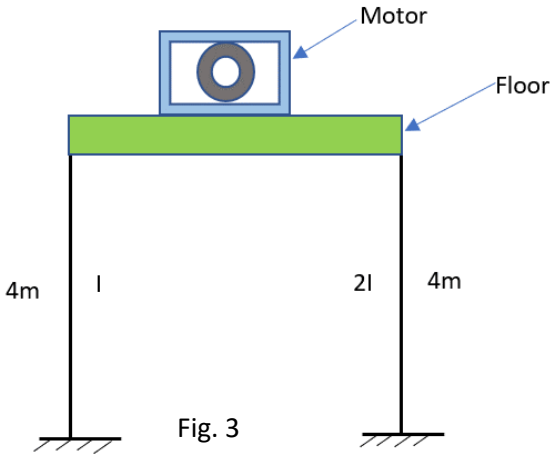

Motor
Floor
4m
I
2I
4m

Fig. 3

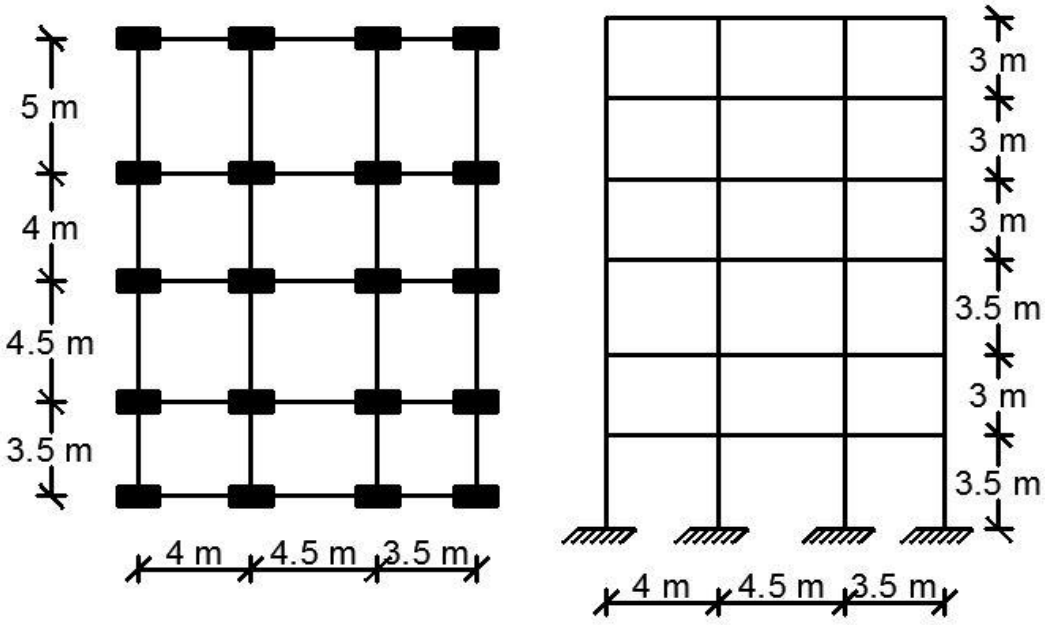

5 m
4 m
4.5 m
3.5 m
4 m
4.5 m
3.5 m
3 m
3 m
3 m
3.5 m
3 m
3.5 m
4 m
4.5 m
3.5 m

Fig. 4